UNIVERSITY OF SCIENCE AND TECHNOLOGY OF HANOI
Department of Information and Communication Technology

BACHELOR THESIS DEFENSE

Hospital Patient Management System

A web application for managing patient records

Trần Hải Nam — 23BI14325

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AGENDA · 15 MINUTES

Presentation Outline

Nội dung trình bày

01	**Problem & Objectives** Vấn đề & mục tiêu	2 min
02	**System Design — architecture & database** Thiết kế hệ thống — kiến trúc & cơ sở dữ liệu	4 min
03	**Key Features — list, change history, validation** Tính năng chính — danh sách, lịch sử thay đổi, kiểm tra dữ liệu	4 min
04	**Live Demo** Trình diễn trực tiếp	2 min
05	**Testing & Coverage** Kiểm thử & độ phủ	2 min
06	**Lessons, Limitations & Future Work** Bài học, hạn chế & hướng phát triển	1 min

INTRODUCTION

Problem Statement

Bài toán đặt ra

SCATTERED

Records live on paper, in stacks

Retrieving one record means flipping through an entire pile of documents.

Hồ sơ giấy, phân tán nhiều nơi

DUPLICATED

Same data copied by hand

Manual copying to multiple places makes retrieval slow, inconsistent — and produces duplicates.

Sao chép thủ công → trùng lặp, sai lệch

UNTRACEABLE

Edits overwrite without a trace

No reliable account of how a record reached its current state.

Sửa là mất dấu vết dữ liệu cũ

→ **One place where records are created, found and edited — and every change is remembered.**

INTRODUCTION

Objectives & Scope

Mục tiêu & phạm vi — 8 yêu cầu chức năng (FR-01…08)

FR-01 … 06

Full record lifecycle

Create, browse, view in detail, edit, delete — plus bulk delete with a confirmed action.

FR-02

A list that stays findable

Paginated list, search by a chosen field, narrowed by status and gender filters.

FR-07

Field-level change history

Every edit records what changed, the old and new value, who made it, and when.

FR-08

Validation at both ends

Every submission checked in the browser and again on the server — invalid data never reaches storage.

CARE STATUS **Waiting** → **Examining** → **Admitted** → **Discharged**

SYSTEM DESIGN

Technology Stack

Công nghệ sử dụng

CLIENT

Angular

Single-page application; bootstrapped from the ng-matero admin scaffold on Material Design.

SERVER

Spring Boot

REST API in Java; JPA / Hibernate for persistence, Jakarta Validation for input rules, built with Maven.

DATABASE

Oracle + Flyway

Schema versioned as migration scripts, applied automatically at startup.

- Credentials live outside the code — read from environment variables at startup, never committed.
- The app runs as a dedicated database account with only the privileges it needs.

SYSTEM DESIGN

System Architecture

Kiến trúc ba tầng, server chia ba lớp

CLIENT

Browser SPA

Owns everything the user sees; reports invalid input as it occurs.

HTTP ⇄ JSON

SERVER — SPRING BOOT

Interface layer	Receives the request, shapes the reply envelope
Business layer	Applies the rules — validation, change tracking
Data-access layer	Reads and writes records

JPA ⇄ SQL

DATABASE

Oracle

Stores the records; Flyway keeps the schema versioned.

→ **A change in one layer leaves the others untouched — and each layer can be tested on its own.**

SYSTEM DESIGN

Database Design

Bệnh nhân là thực thể trung tâm

Profile groups `1 – 1`

Contact, insurance, emergency contact, medical info

Change history `1 – N`

One row per changed field; deleted together with the patient

Treatment episodes `1 – N`

One visit = one self-contained record of diagnoses (ICD-10) and prescriptions

←

CENTRAL ENTITY

PATIENT

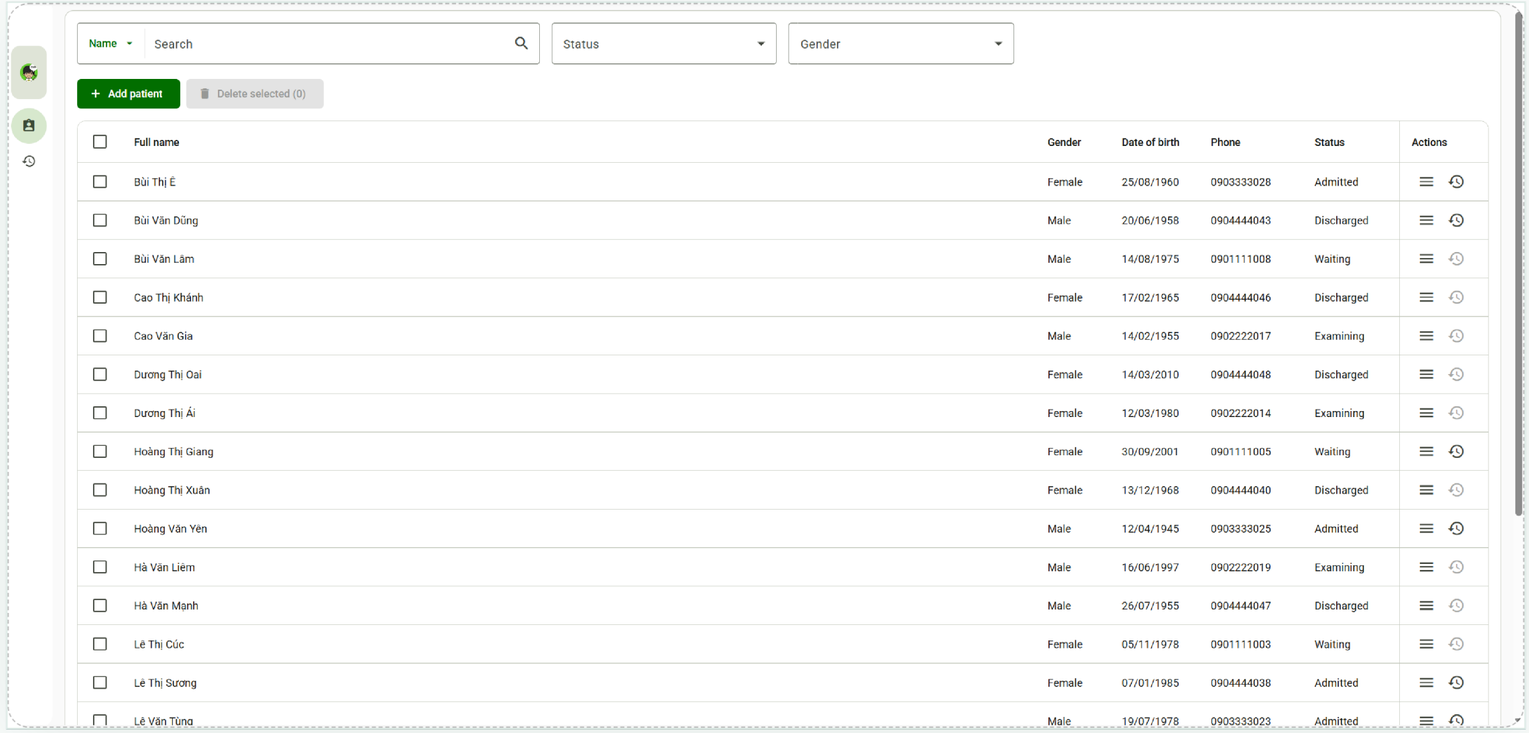
17 editable fields · identified by a displayed PID code

- Status is a coded attribute — one of four care stages
- Reference tables supply names: departments, doctors, medicines, ICD-10 codes
- No history entry can outlive the record it describes

KEY FEATURES

Patient List & Search

Danh sách, tìm kiếm, lọc & phân trang



Search by a chosen field

Name, phone, PID… selected in a dropdown

Status & gender filters

Combine freely with the search term

Server-side pagination

Stays responsive as records grow — one page at a time

Bulk delete, confirmed

Several selected records removed in one confirmed action

→ **All combinations of search, filter and paging are served by a single dynamic query.**

KEY FEATURES

Change History

Lịch sử thay đổi mức từng trường — mỗi lần sửa đều để lại dấu vết

Patient	Changed field	Old value	New value	Changed by	Date	Actions
Lê Văn Tùng	Emergency contact name	Lê Thị Vân	Lê Thị Văn	SYSTEM	17/06/2026	
Lê Thị Sương	Status	Discharged	Admitted	SYSTEM	17/06/2026	
Hoàng Văn Yên	Date of birth	08/03/1945	12/04/1945	SYSTEM	17/06/2026	
Hoàng Thị Giang	Blood type	O+	B+	SYSTEM	17/06/2026	
Bùi Văn Dũng	Address	Số 52, Nguyễn Văn Huyên, Cầu Giấy, Hà Nội	Số 42, Nguyễn Văn Huyên, Cầu Giấy, Hà Nội	SYSTEM	17/06/2026	
Bùi Thị Ê	Email	--	eogibiteo@gmail.com	SYSTEM	12/06/2026	

Items per page: 20 1 - 6 of 6

ONE EDIT, ONE FIELD = **field name** **old value** **new value** **changed by** **changed date**

- The business layer compares old and new values across all 17 editable fields on every update
- Resubmitting unchanged values records nothing

KEY FEATURES

Validation & Error Handling

Kiểm tra hai lớp — phản hồi một khuôn

CHECK 1 · BROWSER

Form rules

Invalid input is reported the moment it occurs, before anything is sent.

→

CHECK 2 · SERVER

Independent validation

Every request is checked in its own right — a 400 reply names each offending field.

→

STORAGE

Only valid data lands

Malformed submissions never reach the database.

ONE RESPONSE ENVELOPE — SUCCESS OR FAILURE

`success` `status code` `message` `data` `fieldErrors` `timestamp`

Internal errors are logged on the server — never leaked to the caller as stack traces.

LIVE DEMO

The system, end to end

Trình diễn trực tiếp — 2 phút

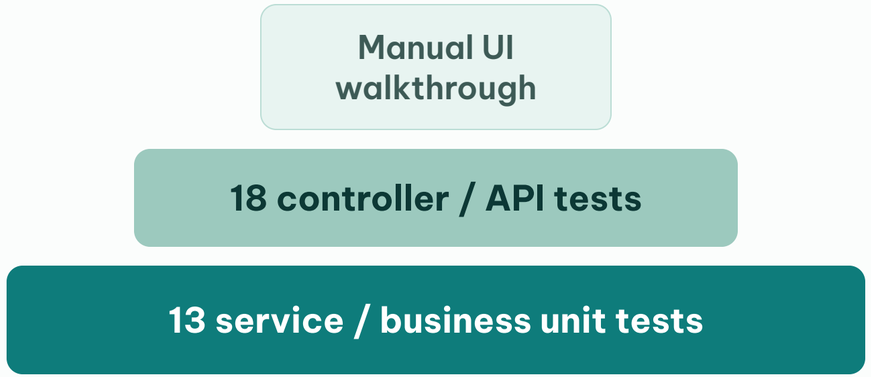
1. Create a patient — watch both validation checks
2. Search and filter the list
3. Edit a record — then open its change history
4. Bulk delete with confirmation

Fallback: screenshots on the previous slides · *Dự phòng: ảnh chụp màn hình ở các slide trước*

EVALUATION

Testing & Coverage

31 test tự động · độ phủ JaCoCo 89%



Each test isolates one layer — the whole suite runs in seconds, no database needed.

89% **overall line coverage** (JaCoCo)



Entry point is bootstrap code — no business logic to test.

CONCLUSION

Lessons Learned

Bài học rút ra

01 Structure comes first

Shaping the layers early kept the project from growing disorganized — and made each rule testable on its own.

02 CRUD is one round trip, not four features

Form → validation → business rules → storage → response. Once that loop was clear, every operation became a variation of it.

03 Consistency pays for itself

One response envelope and one dynamic query for all criteria made the system predictable and easy to manage.

CONCLUSION

Limitations & Future Work

Hạn chế & hướng phát triển

KNOWN LIMITATIONS

- Single user category, no roles — suits a trusted setting
- Status transitions are not constrained in order
- No guard against two users editing the same record
- Statistics page still under development
- Out of scope: billing, insurance claims, record exchange, mobile app

NEXT STEPS

- **Login and per-role permissions**
- **Manage reference lists — medicines, doctors, diseases**
- **Let doctors add diagnoses and prescriptions in the UI**
- **Constrain the order of care-status transitions**
- **Finish statistics and export reports to PDF**

QUESTIONS & ANSWERS

Thank you

Em xin cảm ơn hội đồng — sẵn sàng nhận câu hỏi

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